The Standard Model on the world brane of 7+1 spacetime: provenance

Sources, data, decisions, derivations and the verification record of the Standard Model on the brane, the portals of dirac16complex and neutrino flavour conversion — sections 17–19 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
Document generated by tools/make_docs.py on 2026-10-27

Abstract

This record states what was asked, which experimental data were used and where they come from, every decision taken with the equation behind it, how each result was derived and how it was checked: the 23 checks of sections 17–19 of 7+1.nb (23 True), the 32 boolean checks of the independent Python program tools/ref_sm.py (32 True), the comparison of every number printed by both programs, and the SHA-256 hashes of the files.

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1. The task

This stage of the project asked (summarized; the original wording is the author's private working material) for the TOTALITY of the Standard Model together with gravity and the field dirac16complex on the 7+1 spacetime of this project (one time coordinate x_0, seven space coordinates $x_1, \ldots, x_7$), following the ideas of the repository github.com/once-ere/Dirac_claude — there the Standard Model lives on a seven-dimensional fold of a 4+4 spacetime — but not its metric. Here the Standard Model lives on the 3+1-dimensional world brane $x_4 = \cdots = x_7 = 0$, the auxiliary coordinates form a torus, and gravity and dirac16complex live in the whole spacetime. Every documentation file has a companion provenance file and a symbol file, in Markdown and LaTeX with PDFs; no document refers the reader to another one, and every equation used is written in the document that uses it.

2. Data and their sources

quantity	value used	source
G_F	1.1663788×10^{-5} GeV^{-2}	PDG 2025
m_W, m_Z, m_H	80.3692, 91.1880, 125.20 GeV	PDG 2025
CKM: s_{12}, s_{13}, s_{23}, δ	0.22501, 0.003732, 0.04183, 1.147	PDG 2024 (CKM review, global fit)
J, A, $\bar{\rho}$, $\bar{\eta}$ (for comparison)	$(3.12^{+0.13}_{-0.12}) \times 10^{-5}$; 0.826 ± 0.016; 0.1591 ± 0.0094; 0.3523 ± 0.0073	PDG 2024
m_u, m_d, m_s	2.16, 4.70, 93.5 MeV ($\overline{\text{MS}}$, 2 GeV)	PDG 2025

quantity	value used	source
m_c, m_b, m_t	1.2730 GeV, 4.183 GeV ($\overline{\text{MS}}$), 172.56 GeV (direct)	PDG 2025
$\sin^2\theta_{12}$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$, δ	0.307, 0.561, 0.02195, 177°	NuFit-6.0, normal ordering, without SK atmospheric data
Δm^2_{21}, Δm^2_{31}	7.49×10^{-5}, 2.534×10^{-3} eV²	NuFit-6.0 (same set)
H_0, Ω_m, Ω_r	67.4 km/s/Mpc, 0.315, 9.14×10^{-5}	Planck 2018 ($t_0 = 13.787 \pm 0.020$ Gyr for comparison)
M_{Pl} (reduced)	2.435×10^{18} GeV	standard value
$\hbar c$	$1.973269804\times10^{-16}$ GeV m	CODATA
N_A	6.02214076×10^{23} mol^{-1}	SI (exact)
T_γ	2.7255 K	CMB temperature (Fixsen 2009)
k_B	8.617333262×10^{-5} eV/K	SI (exact)
$R_\odot$; $n_e(r)$	6.957×10^8 m; $245 N_A e^{-10.54 r/R_\odot}$ cm^{-3}	IAU nominal radius; exponential fit to the standard solar model (Bahcall 1989)
DUNE: L, E, ρ, Y_e	1300 km, 2.5 GeV, 2.848 g/cm³, 0.5	DUNE design baseline and crust density

Full references: S. Navas et al. (Particle Data Group), Phys. Rev. D 110 (2024) 030001 and the 2025 update (pdg.lbl.gov); I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, T. Schwetz, JHEP 12 (2024) 216 (NuFit-6.0); Planck Collaboration, Astron. Astrophys. 641 (2020) A6; D. J. Fixsen, Astrophys. J. 707 (2009) 916; J. N. Bahcall, *Neutrino Astrophysics* (Cambridge, 1989).

Software: Wolfram Language 15.0.1 (front end WolframNB.exe); Python 3.13.17 with numpy 2.5.3, scipy 1.18.1, sympy 1.14.0.

3. Decisions

#	decision	reason / equation
1	the Standard Model lives on the brane $W = \{y = 0\}$ of a torus T^4 of the auxiliary coordinates	the 7+1 analogue of the fold of Dirac_claude; $\int_W d^4x\sqrt{\lvert h\rvert} = \int d^8X \sqrt{\lvert g\rvert}\,\delta^4(y)/b^4$
2	three generations with right-handed neutrinos (Dirac neutrinos)	anomalies cancel with or without ν_R; ν_R gives Dirac masses $m = yv/\sqrt{2}$
3	tree-level electroweak relations with the on-shell weak angle	$\cos\theta_W = m_W/m_Z$ makes $\rho = m_W^2/(m_Z^2\cos^2\theta_W) = 1$ exactly
4	standard parametrization of both mixing matrices	$U = R_{23}U_{13}(\delta)R_{12}$; Yukawa matrices in the basis where the up-type (charged-lepton) Yukawa is diagonal
5	lightest neutrino massless, normal ordering	the oscillation data fix only Δm^2_{ij}
6	brane decomposition with the Dirac representation and $\rho_4 = i\gamma_0$, $\rho_k = -\gamma_{k-4}$ $(k = 5, 6, 7)$	gives anti-Hermitian ρ_k with $\{\rho_k, \rho_l\} = -2\delta_{kl}$, so $R_k = \gamma_5 \otimes \rho_k$ squares to -1
7	the intertwiner by a Schur average	$\sum_K \Gamma^{(K)} s (R^{(K)})^{-1}$ commutes in the sense $\Gamma_A U = U R_A$ for any s; non-zero for a suitable matrix unit s
8	portals are invented couplings at the brane, separate for the two chiral halves	as in Dirac_claude; they are the only contact between dirac16complex and the Standard Model besides gravity
9	neutrino evolution in matter by the exponential of the constant-density Hamiltonian; the Sun by direct integration of the two-flavour equation	exact for constant density; checked against the adiabatic formula
10	the solar two-flavour Hamiltonian uses the effective potential $c_{13}^2 V_{CC}$ and $P_{ee} = c_{13}^4 P_{ee}^{(2)} + s_{13}^4$	standard reduction of three flavours when $\Delta m^2_{31} \gg 2EV$

4. Derivations

Anomaly sums (one generation, left-handed +, right-handed −):

$$A_{SU(3)^2U(1)} = 2\cdot\tfrac{1}{6} - \tfrac{2}{3} + \tfrac{1}{3} = 0, \qquad A_{SU(2)^2U(1)} = 3\cdot\tfrac{1}{6} - \tfrac{1}{2} = 0, \qquad A_{SU(3)^3} = 2 - 1 - 1 = 0,$$
$$A_{U(1)^3} = 6\big(\tfrac{1}{6}\big)^3 - 3\big(\tfrac{2}{3}\big)^3 - 3\big(-\tfrac{1}{3}\big)^3 + 2\big(-\tfrac{1}{2}\big)^3 - (-1)^3 = \tfrac{1}{36} - \tfrac{8}{9} + \tfrac{1}{9} - \tfrac{1}{4} + 1 = 0,$$
$$A_{\mathrm{grav}^2\mathrm{U}(1)} = 6\cdot\tfrac{1}{6} - 3\cdot\tfrac{2}{3} - 3\cdot\big(-\tfrac{1}{3}\big) + 2\cdot\big(-\tfrac{1}{2}\big) - (-1) = 0.$$

Gauge-boson masses. With $\langle\Phi\rangle = (0, v/\sqrt{2})$:

$$\big|D_\mu\langle\Phi\rangle\big|^2 = \frac{v^2}{8}\Big[g^2\big((W^1)^2 + (W^2)^2\big) + \big(gW^3 - g'B\big)^2\Big],$$

whose Hessian in (W^1, W^2, W^3, B) has the eigenvalues $g^2v^2/4$ (twice), $(g^2+g'^2)v^2/4$ and 0.

Brane bilinears. $\Gamma_A U = U R_A$ and $U^\dagger U = 1$ give $\bar\psi\Gamma^\mu\psi' = \psi^\dagger\Gamma_0\Gamma^\mu\psi' = \zeta^\dagger R_0 R^\mu \zeta'$, and $R_0 R^\mu = (\gamma_0\gamma^\mu)\otimes 1_4$ for $\mu = 0,\dots,3$, so the transverse index a is summed: $\sum_a \bar\psi_a\gamma^\mu\psi'_a$.

Kaluza–Klein masses. $X = \gamma_5\otimes\vec\rho\cdot\vec q/b_0$ anticommutes with $\gamma_\mu\otimes 1_4$ (γ_5 anticommutes with γ_μ) and $X^2 = \gamma_5^2\otimes(\vec\rho\cdot\vec q)^2/b_0^2 = -|\vec q|^2/b_0^2$; then $(\gamma\cdot p\otimes 1 + X)^2 = (p^2 - |\vec q|^2/b_0^2)\,1$, so the operator $\gamma\cdot p\otimes 1 + X - m$ is singular exactly when $p^2 = m^2 + |\vec q|^2/b_0^2$.

Non-unitary mixing. For $\mathcal{M} = \begin{pmatrix} m_D & \mu \\ 0 & m1_4 \end{pmatrix}$ with $m \gg \mu, m_D$ the left singular vectors give, to second order in μ/m, $N = (1 - \tfrac{1}{2}\mu\mu^\dagger/m^2)U$, hence $1 - NN^\dagger = \mu\mu^\dagger/m^2$.

Vacuum oscillations. Expanding $|\sum_i U_{\beta i}U^*_{\alpha i}e^{-im_i^2L/(2E)}|^2$ and using unitarity, $\sum_i U_{\beta i}U^*_{\alpha i} = \delta_{\alpha\beta}$:

$$P(\alpha\to\beta) = \delta_{\alpha\beta} - 4\sum_{i>j}\mathrm{Re}\,X_{ij}\sin^2\Delta_{ij} + 2\sum_{i>j}\mathrm{Im}\,X_{ij}\sin 2\Delta_{ij}, \qquad X_{ij} = U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j}.$$

(An earlier version of the notebook used $X_{ij} = U_{\alpha i}U^*_{\beta i}U^*_{\alpha j}U_{\beta j}$, the complex conjugate, which flips the sign of the CP-odd term; it was corrected, and the check compares the formula with the direct evolution for all nine (α,β).)

MSW angle. Diagonalizing $\frac{\Delta m^2}{4E}\begin{pmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix} + \mathrm{diag}(V,0)$ gives $\tan 2\theta_m = \sin 2\theta/(\cos 2\theta - 2EV/\Delta m^2)$.

5. Verification

Evaluation of 7+1.nb: Wolfram 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026); finished Tue 27 Oct 2026 14:15:35; message cells 0. The checks of sections 17–19:

#	check	result	section
1	anomaly cancellation per generation (with nu_R): SU(3)^2U(1), SU(2)^2U(1), U(1)^3, grav^2U(1), SU(3)^3 all vanish; doublets per generation even	True	17.1
2	electric charges Q = T3 + Y: (u, d) = (2/3, -1/3), (nu, e) = (0, -1), u_R 2/3, d_R -1/3, e_R -1, nu_R 0	True	17.1
3	g* = 28 + (7/8) 90 = 106.75 for the Standard Model fields (without nu_R)	True	17.1
4	gauge-boson mass matrix: eigenvalues g^2 v^2/4 (twice), (g^2 + g'^2) v^2/4 and 0 (photon)	True	17.3
5	tree level: m_W == g v/2, m_Z == Sqrt[g^2 + g'^2] v/2, rho == m_W^2/(m_Z^2 cos^2 theta_W) == 1, m_H^2 == 2 lambda v^2	True	17.3
6	CKM: unitary; Jarlskog invariant J = 3.12 x 10^-5 (PDG 2024: 3.12 +0.13 -0.12 x 10^-5); A, rhobar, etabar within the PDG uncertainties	True	17.4
7	Yukawa matrices: singular values x v/Sqrt[2] are the quark masses and |U_uL^dag.U_dL| == |V_CKM|	True	17.4
8	PMNS unitary; Dirac neutrino Yukawa Sqrt[2] U.diag(m)/v reproduces the masses (singular values)	True	17.4

#	check	result	section
9	age of the universe from the brane Friedmann equation: 13.79 Gyr (Planck 2018: 13.787 +- 0.020 Gyr)	True	17.5
10	ADD relation: kappa8 rho4/V_perp == 8 Pi G4 rho4 with G4 = G8/V_perp (kappa8 = 8 Pi G8)	True	17.5
11	brane representation: R_A R_B + R_B R_A == 2 eta17_AB ID16 for R = (gamma_mu (x) 1_4, gamma5 (x) rho_k)	True	18.
12	intertwiner: Gamma17[A].U == U.R_A for A = 0..7, and U is unitary	True	18.
13	chi17 == U.(gamma5 (x) rho4567).U^dag; rho4567 Hermitian with eigenvalues {-1, -1, 1, 1}	True	18.
14	bilinears at the brane: psibar psi' == Sum_a psibar_a psi'_a and psibar Gamma^mu psi' == Sum_a psibar_a gamma^mu psi'_a (mu = 0..3)	True	18.
15	KK operator: X.X == -|q|^2/b0^2 ID16 and X anticommutes with gamma_mu (x) 1_4	True	18.1
16	KK masses: Det[gamma.p (x) 1 + X - m] vanishes on p0^2 - |p|^2 == m^2 + |q|^2/b0^2 (40-digit arithmetic) and not off shell	True	18.1
17	portal mixing: 1 - N.N^dag == mu.mu^dag/m^2 to second order (non-unitary light mixing)	True	18.2
18	vacuum: P(a->b) doubly stochastic; equals delta - 4 Sum Re X sin^2 Delta + 2 Sum Im X sin 2Delta; CPT P(a->b) == P(bbar->abar)	True	19.
19	V_CC(rho Y_e = 1 g/cm^3) == 7.63 x 10^-14 eV; matter evolution doubly stochastic; a flavour-universal potential has no effect	True	19.
20	two-flavour MSW: sin^2 theta_m from diagonalization == formula, and sin^2 2theta_m == 1 at 2 E V == dm^2 cos 2theta	True	19.
21	Sun: numerical P_ee == adiabatic MSW formula to 2 x 10^-3 at 0.3, 1, 5 and 10 MeV; P_ee(0.3 MeV) within 0.03 of the vacuum average; P_ee falls with energy	True	19.
22	cosmic neutrino background: T_nu == 1.945 K and n == 112 cm^-3 per flavour	True	19.
23	dirac16complex portal: non-unitary oscillations, Sum_b P(a->b) < 1 (zero-distance effect); decoupled limit == three-flavour vacuum result	True	19.

The independent Python program tools/ref_sm.py — its boolean checks:

#	check	result
1	anomaly cancellation in each generation (with nu_R): SU(3)^2U(1), SU(2)^2U(1), U(1)^3, grav^2U(1), SU(3)^3 all 0	True
2	Witten SU(2) anomaly: number of doublets per generation (3 colours of Q + 1 L) = 4, even	True
3	electric charges Q = T3 + Y: Q -> (2/3, -1/3), u_R 2/3, d_R -1/3, L -> (0, -1), e_R -1, nu_R 0	True
4	effective number of relativistic degrees of freedom g* = 28 + (7/8) 90 = 106.75 (106.75 + 5.25 if nu_R were thermal)	True
5	gauge-boson mass matrix from |D<Phi>|^2: eigenvalues g^2v^2/4 (twice, W+-), (g^2+g'^2)v^2/4 (Z), 0 (photon)	True
6	tree-level relations: m_W = g v/2, m_Z = sqrt(g^2+g'^2) v/2, rho = m_W^2/(m_Z^2 cos^2) = 1, m_H^2 = 2 lambda v^2	True
7	CKM unitarity V V^dag = 1	True
8	J = c12 c23 c13^2 s12 s23 s13 sin(delta) = 3.12e-5 (PDG 2024: 3.12 +0.13 -0.12 e-5)	True
9	Wolfenstein parameters agree with the PDG fit (A 0.826, rhobar 0.1591, etabar 0.3523) within their quoted uncertainties	True
10	Yukawa matrices Y_u = sqrt2 diag(m_u,m_c,m_t)/v, Y_d = sqrt2 V diag(m_d,m_s,m_b)/v: singular values x v/sqrt2 = masses, |U_uL^dag U_dL| = |V|	True
11	PMNS unitarity	True
12	neutrino Yukawa matrix Y_nu = sqrt2 U diag(m1,m2,m3)/v: masses and |U| recovered (singular value decomposition)	True
13	vacuum: probabilities P(a -> b) form a doubly stochastic matrix (rows and columns sum to 1)	True
14	vacuum: amplitude result = delta_ab - 4 sum_{i>j} Re(X_ij) sin^2 Delta_ij + 2 sum_{i>j} Im(X_ij) sin 2Delta_ij, X_ij = U_ai* U_bi U_aj U_bj*	True
15	CPT: P(a->b) = P(bbar->abar); CP asymmetry P(mu->e) - P(mubar->ebar) proportional to J (vanishes for delta = 0, pi)	True
16	V_CC(rho Y_e = 1 g/cm^3) = 7.63e-14 eV	True
17	matter (constant density): unitary evolution, probabilities doubly stochastic	True
18	matter: a flavour-universal potential has no effect (only differences of potentials matter)	True
19	two-flavour MSW: sin^2 theta_m from diagonalization = formula; maximal mixing (sin^2 2theta_m = 1) at 2 E V = dm^2 cos 2theta	True
20	Sun: numerical P_ee = adiabatic MSW formula to 2e-3 at E = 0.3, 1, 5, 10 MeV	True
21	Sun: P_ee falls from near the vacuum average c13^4 (1 - sin^2(2 theta12)/2) + s13^4 (within 0.03 at 0.3 MeV) to about 0.31 at 10 MeV	True
22	CNB: T_nu = 1.945 K and n = 112 cm^-3 per flavour	True

#	check	result
23	brane representation: gamma_mu (x) 1_4 (mu = 0..3) and gamma5 (x) rho_k (k = 4..7) satisfy {R_A, R_B} = 2 eta_AB	True
24	intertwiner U (Schur average over the 256 Clifford products): Gamma17[A] U = U R_A for A = 0..7, U unitary	True
25	8D chirality at the brane: chi17 = U (gamma5 (x) rho4567) U^dag, rho4567 Hermitian with eigenvalues +1, +1, -1, -1	True
26	bilinears at the brane: Psibar Psi' = sum_a psibar_a psi'_a and Psibar Gamma^mu Psi' = sum_a psibar_a gamma^mu psi'_a (mu = 0..3)	True
27	Kaluza-Klein: the 16x16 operator gamma.p (x) 1 + gamma5 (x) rho.q/b0 - m is singular exactly on p0^2 - |p|^2 = m^2 + |q|^2/b0^2	True
28	KK operator: X^2 = -|q|^2/b0^2 and X anticommutes with gamma^mu (x) 1	True
29	non-unitary oscillations: sum_b P(a->b) = (N N^dag)_aa^2-type deficit < 1 (zero-distance effect), complemented by the heavy states	True
30	portal decoupling: for mu -> 0 the light block becomes the PMNS matrix and the oscillation probabilities the three-flavour vacuum ones (section 19)	True
31	age 13.79 Gyr (Planck 2018: 13.787 +- 0.020)	True
32	ADD relation: 3 Ha^2 = kappa8 rho4/V_perp = 8 pi G4 rho4 with G4 = G8/V_perp	True

Numbers printed by both programs (relative differences; the Python program prints $|V|$ with 6 and $|U|$ with 5 decimals):

quantity	values	max. relative difference
$v, g, g', \sin^2\theta_W, e, 1/\alpha, \lambda$	7	2.1×10^{-16}
$\lvert V_{\rm CKM}\rvert$	9	2.3×10^{-5}
$J_{\rm CKM}$	1	2.2×10^{-16}
$\bar\rho, \bar\eta$	2	3.5×10^{-16}
$\lvert U_{\rm PMNS}\rvert$	9	3.2×10^{-5}
$J_{\rm PMNS}$	1	1.2×10^{-16}
m_1, m_2, m_3 (eV)	3	0
y_t	1	1.1×10^{-16}
age (Gyr)	1	1.0×10^{-15}
$V_\perp$, $V_\perp^{1/4}$, lowest KK mass	3	2.2×10^{-16}
light (eV) and heavy (GeV) masses with the portal	7	9.9×10^{-7}
$(1 - NN^\dagger)_{ee}$	1	3.3×10^{-7}
DUNE vacuum $P(\mu \to e)$, $P(\bar\mu \to \bar e)$, $P(\mu \to \mu)$	3	4.3×10^{-16}
DUNE matter $P(\mu \to e)$, $P(\bar\mu \to \bar e)$	2	1.2×10^{-15}
vector portal: $P(\mu \to e)$ with, without	2	9.0×10^{-16}
Sun P_{ee} (E, numerical, adiabatic)	12	3.8×10^{-6}
supernova resonance densities H, L	2	1.9×10^{-16}
T_ν, n_ν	2	1.3×10^{-16}

SHA-256 of the files (at the time this document was generated):

```
7+1.nb                          060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5
    ...d3a
tools/make_7p1_nb.py            2fac94dfd0c9fcf3aed8d6e90a77fbbb8ab1c8450819ff591b7e5cf7b5787
    ...d6f
tools/nb_part_gravity.py        55d16f735e16df7a7cd2aa70d322969bc6cf82c777965f112b5a9acf46a6a
    ...d5f
tools/nb_part_ks_gravity.py     886814fedc4cd015c63b950ee83724025ba21fd6c3e56b68dcb7f2f07afdc
    ...205
tools/nb_part_sm.py             20782f1fe557fd3f3dea7417e89ecd42a1602f7a48599fcdcfd0cff77f120
    ...ecc
tools/lovelock71.py             d93ecbd0f82adb972218d59dfe32999a4af944945ca9758607f3a633e1af9
    ...5a1
tools/ref_gravity.py            2bfdc087ed4dc7d84258e04449d11141b415f54b992e65d42fa42cace08ff
    ...6bd
```

```
tools/ref_ks_gravity.py            8d082985c5161254d7c6294c0310f7807aa037bff27a8cd3ec4d3f8a9f6e0
    ...550
tools/ref_sm.py                    d36d23f472c8339d80f1d0b1eb46c509398562053ead7033d96b054782e73
    ...b49
tools/ref71.py                     4ac783b1ab221317387f2a74e0be60f114fd64940fe3ea96af79b33ee7dcb
    ...8b0
tools/ref44.py                     ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064f
    ...af5
```

Output of tools/ref_sm.py (independent checks of sections 17-19; exit code 0) (complete):

```
anomaly cancellation in each generation (with nu_R): SU(3)^2U(1), SU(2)^2U(1), U(1)^3,
    ...grav^2U(1), SU(3)^3 all 0 : True
Witten SU(2) anomaly: number of doublets per generation (3 colours of Q + 1 L) = 4, even : True
electric charges Q = T3 + Y: Q -> (2/3, -1/3), u_R 2/3, d_R -1/3, L -> (0, -1), e_R -1, nu_R 0
    ...: True
effective number of relativistic degrees of freedom g* = 28 + (7/8) 90 = 106.75 (106.75 + 5.25
    ...if nu_R were thermal) : True
gauge-boson mass matrix from |D<Phi>|^2: eigenvalues g^2v^2/4 (twice, W+-), (g^2+g'^2)v^2/4
    ...(Z), 0 (photon) : True
NUM v = (sqrt2 G_F)^(-1/2) [GeV], g, g', sin^2 theta_W (on shell), e, 1/alpha(tree), lambda =
    ...m_H^2/(2v^2) : (246.21964023926205, 0.6528252573344827, 0.34994623401223707,
    ...0.22320949284344327, 0.3084276316025824, 132.10014858375337, 0.1292805756612353)
tree-level relations: m_W = g v/2, m_Z = sqrt(g^2+g'^2) v/2, rho = m_W^2/(m_Z^2 cos^2) = 1,
    ...m_H^2 = 2 lambda v^2 : True
CKM unitarity V V^dag = 1 : True
NUM CKM |V| (rows ud us ub / cd cs cb / td ts tb) : [[0.97435, 0.225008, 0.003732], [0.224876,
    ...0.973489, 0.04183], [0.008583, 0.041109, 0.999118]]
NUM Jarlskog J = Im(V_us V_cb V_ub* V_cs*) : 3.1169935287554706e-05
J = c12 c23 c13^2 s12 s23 s13 sin(delta) = 3.12e-5 (PDG 2024: 3.12 +0.13 -0.12 e-5) : True
NUM Wolfenstein lambda = s12, A = s23/s12^2, rhobar + i etabar = -V_ud V_ub*/(V_cd V_cb*) :
    ...(0.22501, 0.8261981634695225, np.float64(0.15918443594247167),
    ...np.float64(0.35227461890586964))
Wolfenstein parameters agree with the PDG fit (A 0.826, rhobar 0.1591, etabar 0.3523) within
    ...their quoted uncertainties : True
Yukawa matrices Y_u = sqrt2 diag(m_u,m_c,m_t)/v, Y_d = sqrt2 V diag(m_d,m_s,m_b)/v: singular
    ...values x v/sqrt2 = masses, |U_uL^dag U_dL| = |V| : True
NUM top Yukawa y_t = sqrt2 m_t/v : 0.9911341438317451
PMNS unitarity : True
NUM PMNS |U| and J_PMNS : ([[0.82328, 0.54796, 0.14816], [0.27491, 0.61298, 0.74073], [0.49663,
    ...0.56921, 0.65526]], np.float64(0.0017359130944756358))
NUM Dirac neutrino masses (lightest m1 = 0, eV) and their sum : (0.0, 0.008654478609367522,
    ...0.05033885179461287, 0.058993330403980394)
neutrino Yukawa matrix Y_nu = sqrt2 U diag(m1,m2,m3)/v: masses and |U| recovered (singular
    ...value decomposition) : True
NUM largest neutrino Yukawa sqrt2 m3/v : 2.891316340688847e-13
vacuum: probabilities P(a -> b) form a doubly stochastic matrix (rows and columns sum to 1) :
    ...True
vacuum: amplitude result = delta_ab - 4 sum_{i>j} Re(X_ij) sin^2 Delta_ij + 2 sum_{i>j}
    ...Im(X_ij) sin 2Delta_ij, X_ij = U_ai* U_bi U_aj U_bj* : True
NUM DUNE vacuum (L = 1300 km, E = 2.5 GeV): P(mu->e), P(mubar->ebar), P(mu->mu) :
    ...(np.float64(0.04883685296656589), np.float64(0.05019851034574761),
    ...np.float64(0.012813139189347619))
CPT: P(a->b) = P(bbar->abar); CP asymmetry P(mu->e) - P(mubar->ebar) proportional to J
    ...(vanishes for delta = 0, pi) : True
NUM V_CC = sqrt2 G_F n_e for rho Y_e = 1 g/cm^3 [eV] (textbook 7.63e-14 eV) :
    ...7.632466872241316e-14
V_CC(rho Y_e = 1 g/cm^3) = 7.63e-14 eV : True
matter (constant density): unitary evolution, probabilities doubly stochastic : True
NUM DUNE in matter (rho = 2.848 g/cm^3, Y_e = 0.5): P(mu->e), P(mubar->ebar) :
    ...(np.float64(0.07232693782692827), np.float64(0.029191799469459462))
matter: a flavour-universal potential has no effect (only differences of potentials matter) :
    ...True
```

```
two-flavour MSW: sin^2 theta_m from diagonalization = formula; maximal mixing (sin^2 2theta_m =
    ...1) at 2 E V = dm^2 cos 2theta : True
NUM Sun: P_ee (numerical through the solar density profile, adiabatic formula) for E = 0.3, 1,
    ...5, 10 MeV : {0.3: (np.float64(0.5251624291065616), 0.5251645055880894), 1.0:
    ...(np.float64(0.4584114174121251), 0.4584122490661078), 5.0:
    ...(np.float64(0.3103377102301538), 0.3103335111666083), 10.0:
    ...(np.float64(0.2979217906988696), 0.29791748878645763)}
Sun: numerical P_ee = adiabatic MSW formula to 2e-3 at E = 0.3, 1, 5, 10 MeV : True
Sun: P_ee falls from near the vacuum average c13^4 (1 - sin^2(2 theta12)/2) + s13^4 (within
    ...0.03 at 0.3 MeV) to about 0.31 at 10 MeV : True
NUM supernova MSW resonance densities for E = 20 MeV, Y_e = 0.5: H (dm31, theta13), L (dm21,
    ...theta12) [g/cm^3] : (1587.1391520947036, 18.939748107619376)
NUM cosmic neutrino background: T_nu = (4/11)^(1/3) T_gamma [K], number density per flavour
    ...[cm^-3] : (1.9453688391750839, 112.01641316574336)
CNB: T_nu = 1.945 K and n = 112 cm^-3 per flavour : True
brane representation: gamma_mu (x) 1_4 (mu = 0..3) and gamma5 (x) rho_k (k = 4..7) satisfy
    ...{R_A, R_B} = 2 eta_AB : True
intertwiner U (Schur average over the 256 Clifford products): Gamma17[A] U = U R_A for A =
    ...0..7, U unitary : True
8D chirality at the brane: chi17 = U (gamma5 (x) rho4567) U^dag, rho4567 Hermitian with
    ...eigenvalues +1, +1, -1, -1 : True
bilinears at the brane: Psibar Psi' = sum_a psibar_a psi'_a and Psibar Gamma^mu Psi' = sum_a
    ...psibar_a gamma^mu psi'_a (mu = 0..3) : True
Kaluza-Klein: the 16x16 operator gamma.p (x) 1 + gamma5 (x) rho.q/b0 - m is singular exactly on
    ...p0^2 - |p|^2 = m^2 + |q|^2/b0^2 : True
KK operator: X^2 = -|q|^2/b0^2 and X anticommutes with gamma^mu (x) 1 : True
portal mixing: light masses (eV) and heavy masses (GeV) of the 7x7 Dirac mass matrix : ([0.0,
    ...0.008654465673872413, 0.05033886016084743], [1.0, 1.0, 1.00000000025,
    ...1.0000000007499998])
portal mixing: N N^dag = 1 - mu mu^dag/m^2 to second order (non-unitary light mixing), (1 - N
    ...N^dag)_ee : (True, np.float64(9.99999860695766e-10))
non-unitary oscillations: sum_b P(a->b) = (N N^dag)_aa^2-type deficit < 1 (zero-distance
    ...effect), complemented by the heavy states : True
portal decoupling: for mu -> 0 the light block becomes the PMNS matrix and the oscillation
    ...probabilities the three-flavour vacuum ones (section 19) : True
NUM vector portal (V_P,e = 0.2 V_CC): P(mu->e) at DUNE with and without the dirac16complex
    ...potential : (np.float64(0.07704998293140915), np.float64(0.07232693782692827))
NUM age of the universe from the brane Friedmann equation (Planck 2018: H0 = 67.4, Omega_m =
    ...0.315) [Gyr] : 13.790728126409226
age 13.79 Gyr (Planck 2018: 13.787 +- 0.020) : True
ADD relation: 3 Ha^2 = kappa8 rho4/V_perp = 8 pi G4 rho4 with G4 = G8/V_perp : True
NUM ADD example (reduced 8D Planck mass 10 TeV): V_perp [GeV^-4], V_perp^(1/4) [m], lowest KK
    ...mass 2 pi/V_perp^(1/4) [MeV] : (5929225000000.0, 3.079186208511254e-13,
    ...4.026524867292262)
boolean checks: 32   failed: 0
VERDICT: ALL CHECKS PASSED
```